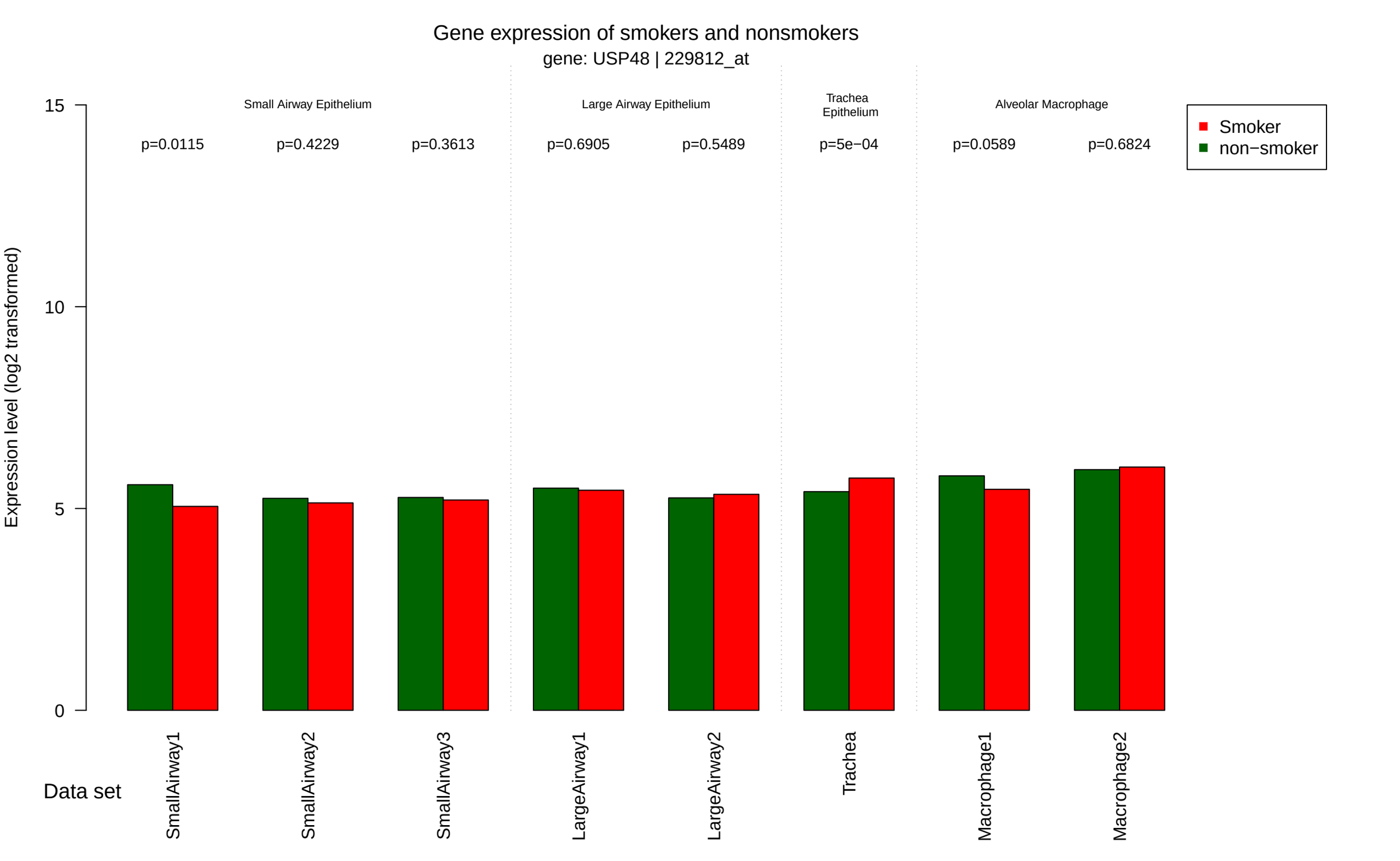
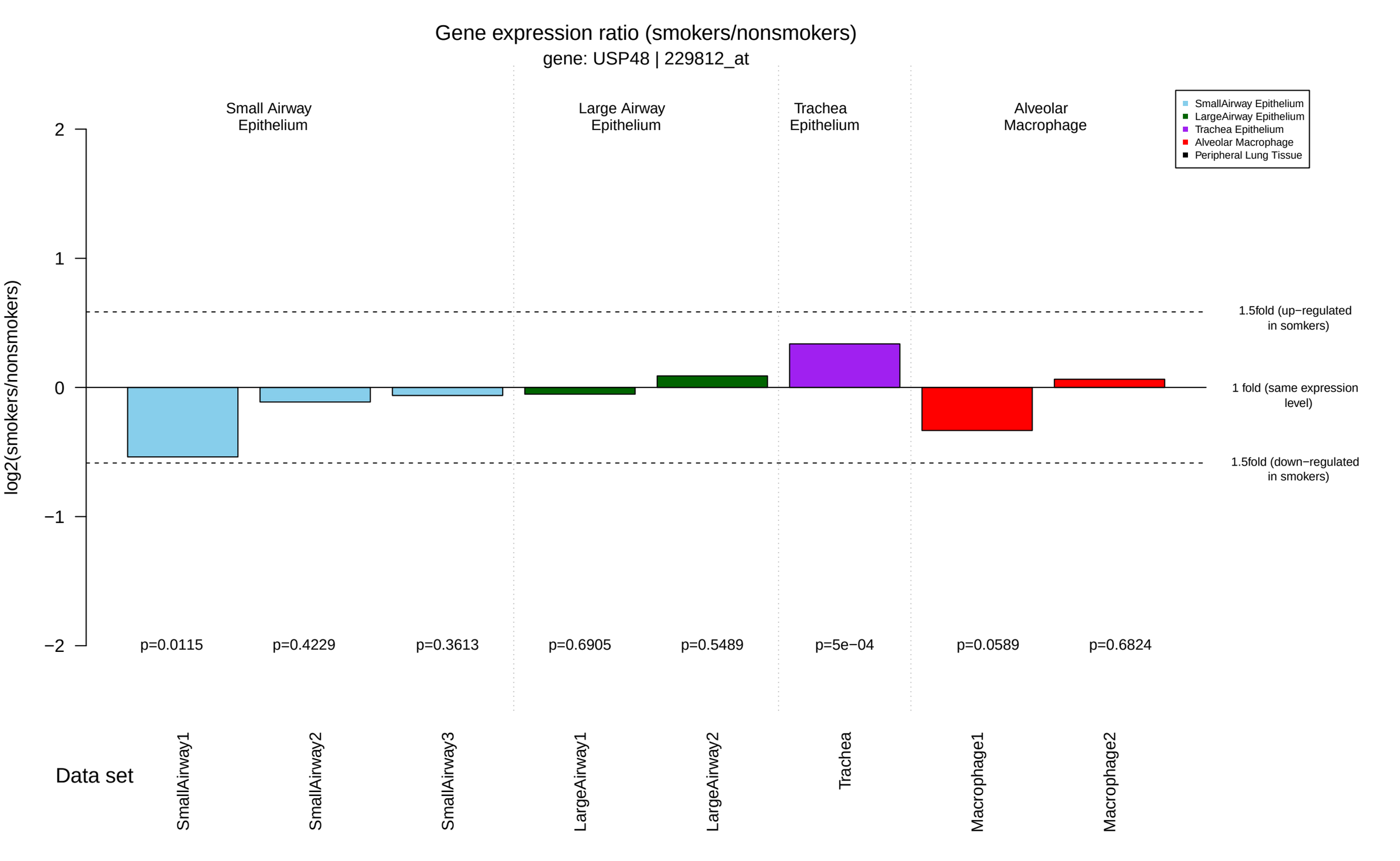
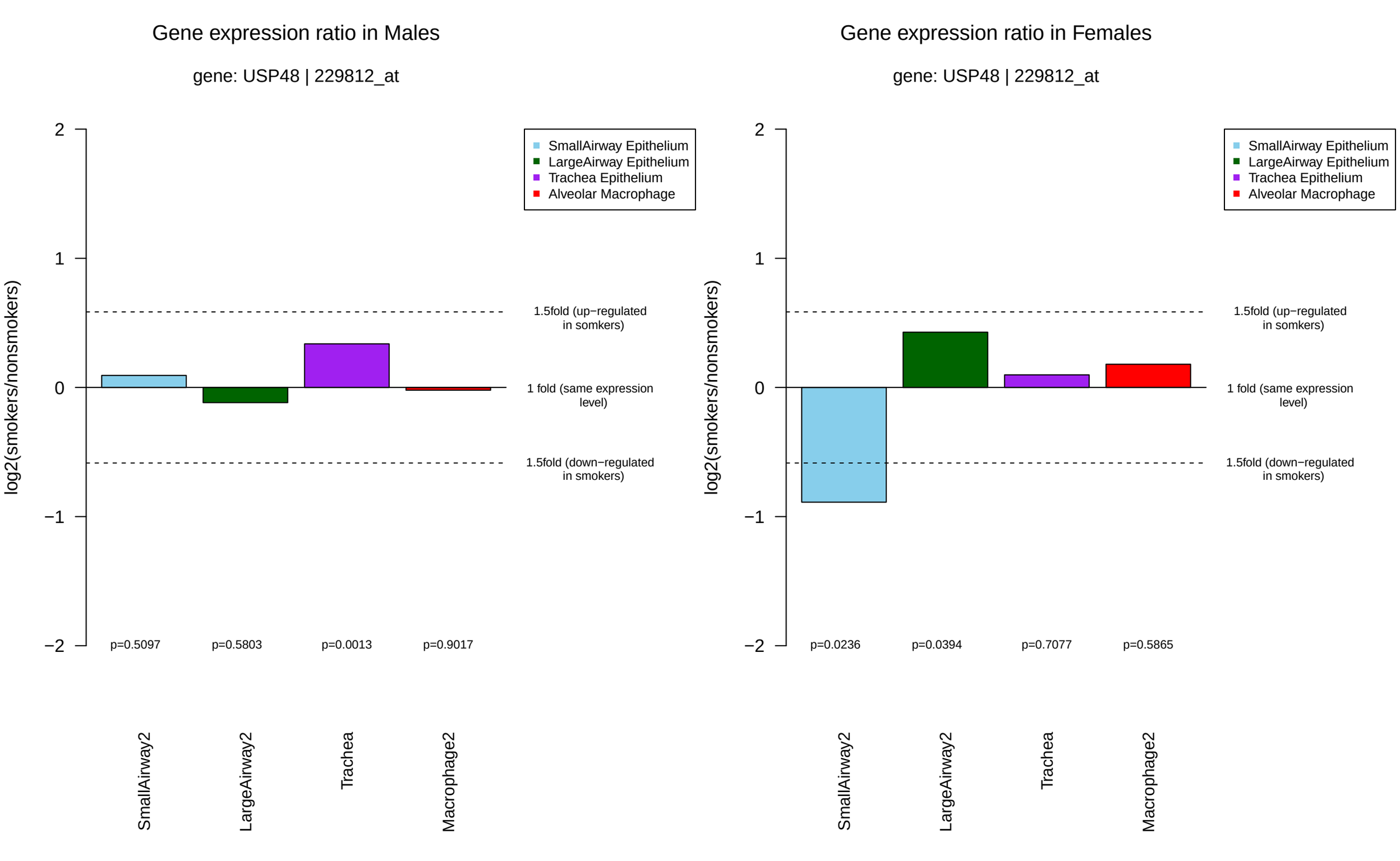
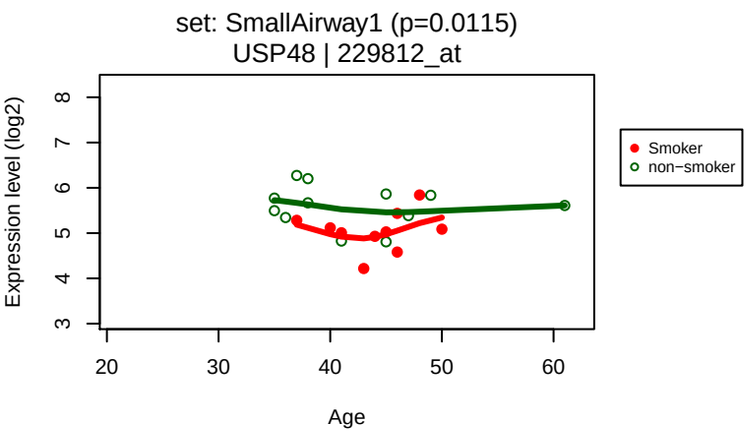
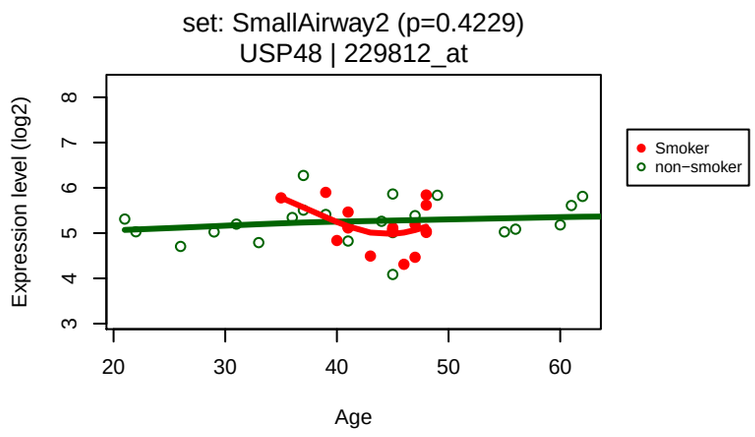
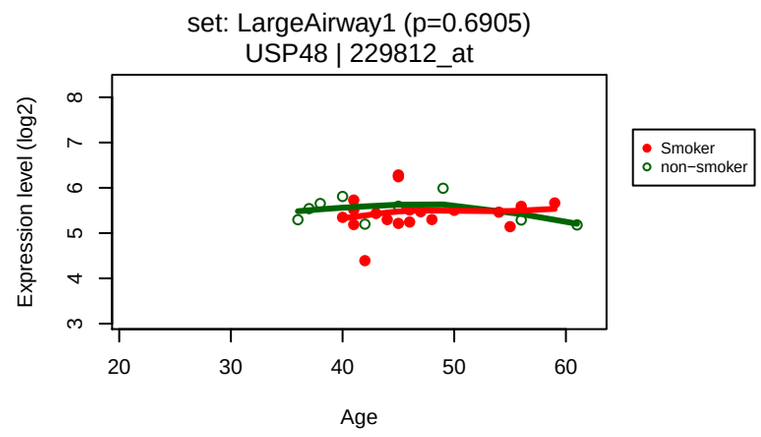

Gene expression of smokers and nonsmokers
gene: USP48 | 229812_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0115
p=0.4229
p=0.3613
p=0.6905
p=0.5489
p=5e−04
p=0.0589
p=0.6824
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

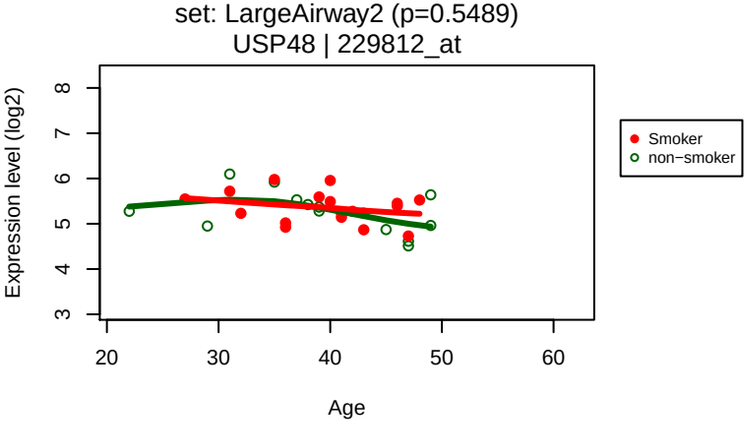

Gene expression ratio (smokers/nonsmokers)
gene: USP48 | 229812_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0115
p=0.4229
p=0.3613
p=0.6905
p=0.5489
p=5e−04
p=0.0589
p=0.6824
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

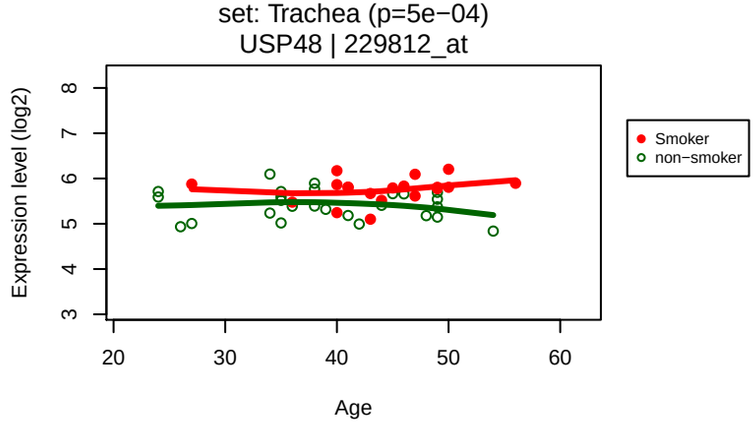

Gene expression ratio in Males
gene: USP48 | 229812_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5097
p=0.5803
p=0.0013
p=0.9017
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: USP48 | 229812_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0236
p=0.0394
p=0.7077
p=0.5865
SmallAirway2
LargeAirway2
Trachea
Macrophage2

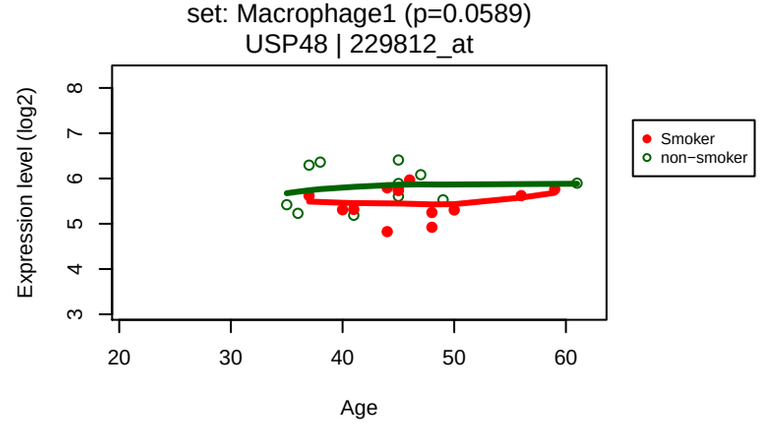

set: SmallAirway1 (p=0.0115)
USP48 | 229812_at
Expression level (log2)
Age
Smoker
non−smoker

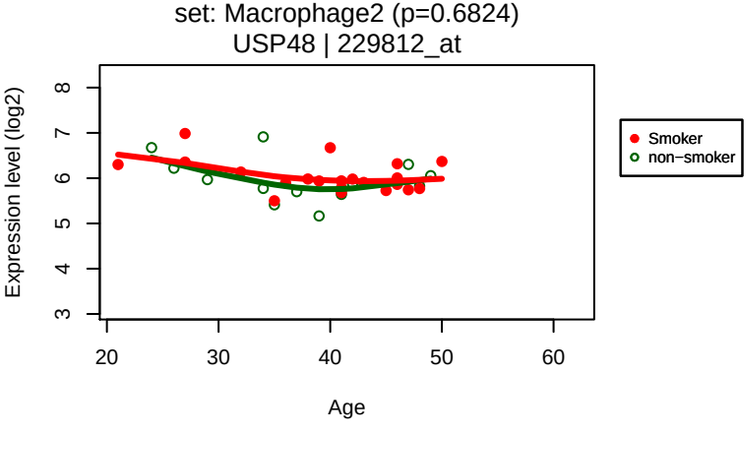

set: SmallAirway2 (p=0.4229)
USP48 | 229812_at
Expression level (log2)
Age
Smoker
non−smoker

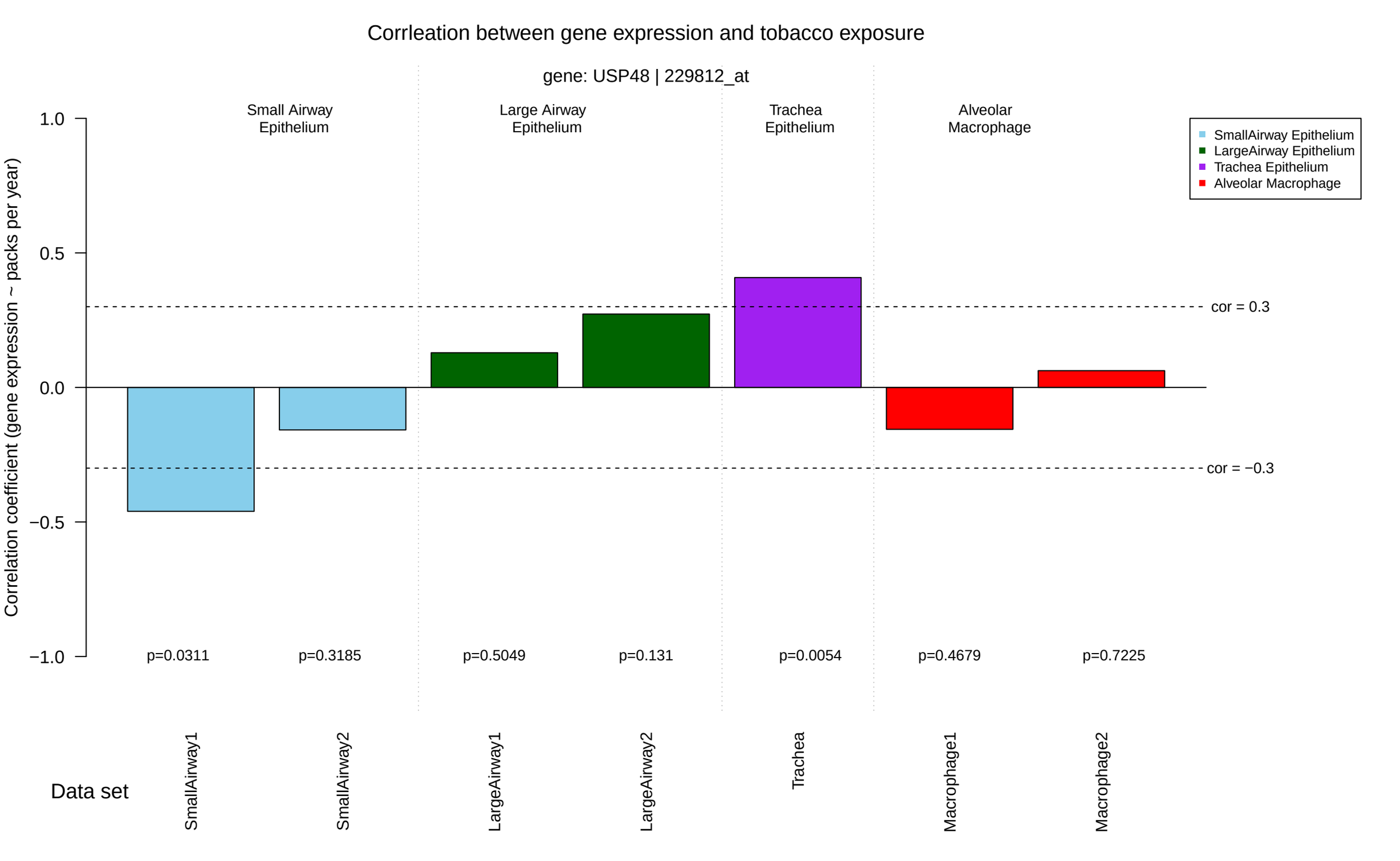

set: LargeAirway1 (p=0.6905)
USP48 | 229812_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5489)
USP48 | 229812_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=5e−04)
USP48 | 229812_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0589)
USP48 | 229812_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6824)
USP48 | 229812_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: USP48 | 229812_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0311
p=0.3185
p=0.5049
p=0.131
p=0.0054
p=0.4679
p=0.7225
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2